

SAADIA ZAKAI

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EDUCATION

University of Maryland
Bachelor of Science, Bioengineering

College Park, MD
Graduated - May 2014

WORK EXPERIENCE

Bioprocess Scale-Up Facility
Lab Technician

College Park, MD
February 2013 – Present

- Assist with upstream and downstream processing in the lab
- Conduct biopharmaceutical projects for industrial and academic clients
- Operate pilot scale fermenters: 10 L, 100 L, and 250 L
- Perform cell separations through continuous centrifugation and filtration
- Employed as a teaching assistant in educational workshops for biotechnology companies and undergraduate classes
- Assist with the operation of chromatography systems and column packing
- Conduct Good Laboratory Practices and Good Documentation Practices
- Revise and follow Standard Operating Procedures to ensure safety in the lab

RESEARCH EXPERIENCE

Bioprocess Scale-Up Facility
Undergraduate Researcher

College Park, MD
Fall 2013

- Independently researched various methods of Hydrophobic Interaction Chromatography (HIC) to increase the efficiency and accuracy for the separation and purification of Green Fluorescent Protein
- Experimented with several different resins to determine the most cost-effective approach
- Tested stepwise and linear gradient techniques to optimize the purification of GFP
- Developed a Standard Operating Procedure (SOP) using a stepwise gradient elution for the BioLogic LP Chromatography System from Bio-Rad

DESIGN EXPERIENCE

Senior Capstone Design Project
Team Member

College Park, MD
August 2013 - May 2014

- Collaborated with team members to design and develop a novel, low-cost respiratory clinical diagnostic device using Laser Speckle Contrast Imaging
- Designed a quantitative assay to measure the biological and mechanical properties of human sputum
- Worked directly with clinical and engineering faculty mentors to determine a correlation between the concentration of neutrophils in sputum samples and absorbance measurements
- Implemented the design of the project using MATLAB and CAD

AWARDS

Student's Choice Award – Senior Capstone Design Competition
Fischell Department of Bioengineering

May 2014

- Awarded by peers and staff in recognition of exceptional work in the design of a clinical diagnostic tool and procedure

SKILLS

- Proficient in Microsoft Office; Experience with MATLAB, C.